Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Expanding and Factorising

Name _____ **Date** _____ No calculator **Version 1** Scan to practise this topic online

KEY WORDS expand *(whakaroha)* · brackets *(taiapa)* · distributive property · factorise *(whakatauwehe)*

A Expanding brackets ●○○

⚠ **Watch out:** Multiply every term inside the bracket: $3(x+2) = 3x+6$, not $3x+2$.

1. Expand.

a) $3(x+2)$ = _____ **b)** $2(x+5)$ = _____ **c)** $4(x-1)$ = _____ **d)** $5(x+3)$ = _____

e) $2(3x+1)$ = _____ **f)** $6(x-2)$ = _____ **g)** $3(2x+5)$ = _____ **h)** $7(x+4)$ = _____

i) $4(3x-2)$ = _____ **j)** $10(x+6)$ = _____ **k)** $5(2x-3)$ = _____ **l)** $8(4x+1)$ = _____

B Brackets with negative numbers ●●○

Remember: A negative number outside changes every sign inside: $-2(x+3) = -2x-6$ and $-2(x-3) = -2x+6$.

2. Expand.

a) $-2(x+3)$ = _____ **b)** $-3(x+1)$ = _____ **c)** $-4(x-2)$ = _____ **d)** $-5(2x+1)$ = _____

e) $-(x+6)$ = _____ **f)** $-2(3x-4)$ = _____ **g)** $-6(x-1)$ = _____ **h)** $-3(2x+5)$ = _____

C Expand and simplify ●●●

Remember: Expand each bracket first, then collect the like terms.

3. Expand and simplify.

a) $2(x+3)+4x$

b) $3(x+2)+2x+5$

c) $5(x-1)-2x$

d) $4(2x+1)+x-6$

e) $2(x+3)+3(x+1)$

f) $3(x+4)+2(x-5)$

g) $4(x+2)+2(2x+1)$

h) $5(x-2)+3(x+4)$

i) $6(x+2)-4x-5$

j) $2(3x+1)+4(x+2)$

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D Factorising ●●○

> **Remember:** Factorising undoes expanding. Find the largest number that divides into every term and write it outside a bracket: $6x + 9 = 3(2x + 3)$.

4. Factorise.

a) $2x + 6$ = _____ **b)** $3x + 12$ = _____ **c)** $5x + 10$ = _____ **d)** $4x - 12$ = _____

e) $4x + 6$ = _____ **f)** $6x - 15$ = _____ **g)** $6x + 12$ = _____ **h)** $7x - 7$ = _____

i) $8x + 20$ = _____ **j)** $15x - 10$ = _____ **k)** $10x + 30$ = _____ **l)** $18x + 9$ = _____

E Fully factorised ●●●

> **Watch out:** Take out the largest common factor. $12x + 18 = 2(6x + 9)$ is only partly factorised: the full answer is $6(2x + 3)$.

5. Factorise fully.

a) $8x + 12$ **b)** $12x + 18$ **c)** $8x + 24$ **d)** $18x - 12$

e) $12x + 24$ **f)** $20x - 30$ **g)** $27x + 36$ **h)** $15x - 30$

i) $24x - 16$ **j)** $24x + 36$ **k)** $20x - 60$ **l)** $28x + 14$

F Problems ●●●

6. A rectangle is 3 cm wide and $x + 4$ cm long. Write an expression for its area without brackets.

7. The area of a rectangle is $6x + 12$ cm². Its width is 6 cm. Factorise the area to find an expression for its length.

8. A rectangle is $x + 3$ cm long and 4 cm wide. Write a simplified expression for its perimeter.

★ Extension

9. Expand and simplify. Take care with the signs.

a) $5(x + 3) - 2(x + 4)$ **b)** $4(2x + 1) - 3(x - 2)$

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1-1 PRIVATE TUTORS

ALGEBRA

Expanding and Factorising

Name _____ **Date** _____ No calculator **Version 2** Scan to practise this topic online

KEY WORDS expand *(whakaroha)* · brackets *(taiapa)* · distributive property · factorise *(whakatauwehe)*

A Expanding brackets ●○○

⚠ **Watch out:** Multiply every term inside the bracket: $3(x+2) = 3x+6$, not $3x+2$.

1. Expand.

a) $2(x+3)$ = _____ **b)** $3(x+4)$ = _____ **c)** $5(x-1)$ = _____ **d)** $4(x+6)$ = _____

e) $3(2x+1)$ = _____ **f)** $7(x-2)$ = _____ **g)** $2(4x+3)$ = _____ **h)** $6(x+5)$ = _____

i) $5(3x-2)$ = _____ **j)** $10(x+7)$ = _____ **k)** $4(2x-5)$ = _____ **l)** $9(3x+1)$ = _____

B Brackets with negative numbers ●●○

Remember: A negative number outside changes every sign inside: $-2(x+3) = -2x-6$ and $-2(x-3) = -2x+6$.

2. Expand.

a) $-3(x+2)$ = _____ **b)** $-2(x+1)$ = _____ **c)** $-5(x-2)$ = _____ **d)** $-4(2x+1)$ = _____

e) $-(x+8)$ = _____ **f)** $-3(2x-5)$ = _____ **g)** $-7(x-1)$ = _____ **h)** $-2(4x+3)$ = _____

C Expand and simplify ●●●

Remember: Expand each bracket first, then collect the like terms.

3. Expand and simplify.

a) $3(x+2)+5x$

b) $2(x+4)+3x+1$

c) $4(x-1)-3x$

d) $5(2x+1)+x-8$

e) $3(x+2)+2(x+1)$

f) $2(x+5)+4(x-3)$

g) $5(x+1)+3(2x+1)$

h) $4(x-3)+2(x+7)$

i) $6(x+3)-2x-9$

j) $2(4x+1)+3(x+2)$

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D Factorising ●●○

Remember: Factorising undoes expanding. Find the largest number that divides into every term and write it outside a bracket: $6x + 9 = 3(2x + 3)$.

4. Factorise.

a) $2x + 10$ = _____ **b)** $3x + 6$ = _____ **c)** $5x + 15$ = _____ **d)** $4x - 20$ = _____

e) $4x + 10$ = _____ **f)** $6x - 3$ = _____ **g)** $6x + 30$ = _____ **h)** $7x - 14$ = _____

i) $8x + 12$ = _____ **j)** $15x - 20$ = _____ **k)** $10x + 70$ = _____ **l)** $18x + 27$ = _____

E Fully factorised ●●●

Watch out: Take out the largest common factor. $12x + 18 = 2(6x + 9)$ is only partly factorised: the full answer is $6(2x + 3)$.

5. Factorise fully.

a) $12x + 8$ **b)** $18x + 12$ **c)** $8x + 40$ **d)** $12x - 30$

e) $12x + 36$ **f)** $30x - 20$ **g)** $18x + 45$ **h)** $15x - 60$

i) $16x - 24$ **j)** $36x + 24$ **k)** $20x - 40$ **l)** $14x + 42$

F Problems ●●●

6. A rectangle is 4 cm wide and $x + 5$ cm long. Write an expression for its area without brackets.

7. The area of a rectangle is $5x + 15$ cm². Its width is 5 cm. Factorise the area to find an expression for its length.

8. A rectangle is $x + 4$ cm long and 4 cm wide. Write a simplified expression for its perimeter.

★ Extension

9. Expand and simplify. Take care with the signs.

a) $6(x + 2) - 3(x + 1)$ **b)** $5(2x + 1) - 2(x - 4)$

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